

TechTrade

Product Definition & Target Audience

1. Define Your Product

1.1 What problem does your product website solve or address?

Many technology consumers, ranging from power users to people who are less technically experienced, go through the same process of deciding what to do with their technology. When a device begins to slow down, experience problems, or become outdated, users may have several options: repair it, upgrade it, sell it, or replace it entirely.

The difficulty is determining which option makes the most financial and practical sense for an individual's specific situation. Consumers may know generally what they want, but technical specifications, product marketing, pricing, and limited knowledge can make it difficult to determine whether purchasing a new device is actually necessary.

TechTrade addresses this problem by helping users make a more informed decision based on their current device, its specifications and value, their personal needs, and their available budget.

1.2 What is your product idea? How does your website support this idea and help solve the problem?

TechTrade is an online analysis and recommendation tool designed to help consumers make better decisions about their technology.

Users provide information about their current device, their needs, and their budget through an analysis form. TechTrade then uses this information to provide a personalized recommendation regarding whether the user should repair, upgrade, sell, or replace their device.

The website supports this idea by providing a simple, user-friendly process that allows consumers to receive useful information without requiring extensive technical knowledge. The goal is to reduce the possibility of making an unnecessary or financially inefficient technology purchase.

TechTrade will provide a no-account experience consisting of a homepage, an analysis input page, and a recommendation output page. The recommendation will include a downloadable PDF, appropriate links, and an option to restart the process.

1.3 What type of website are you building?

TechTrade is an **interactive analysis and recommendation website**.

Rather than functioning as a traditional e-commerce website, TechTrade will simulate the analysis and recommendation process that could eventually become a larger technology decision-making service using hard-coded values for user testing and simulation purposes.

The initial website will be used to determine:

1. Whether the service is effective.
2. Opportunities for future monetization.
3. Preferred UI/UX based on user feedback.

The site will also be developed using an **AGILE development process**, allowing user feedback to influence future versions of the product.

1.4 Mission Statement

TechTrade's mission is to help everyday technology consumers make informed, financially responsible decisions about their devices by analyzing their current technology, personal needs, and budget to determine whether they should repair, upgrade, sell, or replace their device.

TechTrade is designed for both technically experienced users and people who may not be comfortable evaluating technology on their own. By simplifying the decision-making process and presenting recommendations in an intuitive format, TechTrade aims to improve users' technology knowledge, financial decision-making, and overall purchasing experience.

2. Profile of Target Audience

2.1 Primary Target Audience

TechTrade's primary audience is **U.S. technology consumers who are trying to determine what to do with an existing device.**

The target audience includes both technically knowledgeable consumers and people who have limited knowledge of computer hardware and technology purchasing. The common factor among these users is that they want to make a financially responsible decision about an existing device.

The target audience is especially relevant to consumers who are considering one or more of the following:

- Repairing an existing device
- Upgrading an existing device
- Selling an existing device
- Replacing an existing device
- Researching whether purchasing a new device is actually necessary
- Understanding the value and capabilities of their current technology
- Making a technology purchase while staying within a specific budget

2.2 Demographics

Age Range

The target audience is approximately **18–75+ years old.**

TechTrade is intended to be useful across a wide range of ages because the problem it addresses is not limited to a particular generation. A college student with a limited budget may need to determine whether their laptop can be upgraded, while an older adult on a fixed income may need to determine whether replacing a device is financially necessary.

The existing personas demonstrate this range through Alex Martinez, age 21, and Sarah Thompson, age 42. An additional persona, Margaret Grenadine, age 73, expands the audience to include older consumers.

Gender Distribution

TechTrade is intended for **all genders**.

The existing fictional audience includes both male and female users, and the service itself is not designed around a specific gender.

Country

The primary target market is the **United States**.

The existing user stories are based in Rhode Island, and the service is currently designed around U.S. consumers and their technology purchasing decisions.

Urban or Rural Areas

TechTrade is intended for consumers living in **urban, suburban, and rural areas**.

Because the service is web-based, users do not need to live near a physical technology store in order to use it. This allows TechTrade to potentially serve consumers who have limited access to in-person technology advice.

2.3 Socioeconomic Details

Average Income

The target audience is expected to have a **wide range of incomes**, because technology purchasing decisions affect consumers across different socioeconomic groups.

The existing personas range from:

- Alex Martinez — \$18,000/year
- Sarah Thompson — \$78,000/year

Margaret Grenadine represents another important financial situation: an older adult living on a fixed income.

Rather than targeting a single income level, TechTrade is intended to help users determine how their available financial resources should affect their technology decisions.

Education

The target audience may have **anything from a high school education to a college or graduate-level education.**

Technical knowledge is not necessarily connected to formal education, which is why TechTrade is designed to make the analysis process understandable to users who may not have extensive technology experience.

Marital or Family Status

The target audience may be **single, married, partnered, or have children/dependents.**

Family and household responsibilities can influence technology purchasing decisions, particularly when a device represents a significant expense or is needed for work, school, or other responsibilities.

Occupation

TechTrade is intended for a broad range of occupations, including:

- College students
- Part-time employees
- Full-time professionals
- Managers
- Remote workers
- Retirees
- Individuals on fixed incomes
- Other consumers who rely on personal technology

Hours Worked Per Week

The target audience may work anywhere from **0–50+ hours per week**, depending on their occupation and circumstances.

Students and retirees may work fewer hours or not work at all, while full-time professionals may work approximately 40 hours per week or more.

2.4 Web Behavior

How Often Do They Use the Web?

TechTrade's target audience is expected to use the web **regularly, with many users accessing it daily**.

Common activities include:

- Online shopping
- Banking
- Schoolwork
- Work
- Streaming
- Social media
- Technology research
- Product comparison

The existing user stories identify daily web use for both Alex Martinez and Sarah Thompson.

What Devices Do They Use to Access the Web?

Users may access TechTrade through:

- Smartphones
- Laptops
- Desktop computers
- Tablets

The website should therefore use a responsive and accessible design that works across different screen sizes and levels of technical experience.

3. Fictional Target Audience Visitors

Name	Sex	Age	Location	Occupation	Income	Web Use
Alex Martinez	Male	21	Kingston, RI	College Student / Part-Time Retail Employee	\$18,000/year	Daily — schoolwork, shopping, social media, streaming, and technology research
Sarah Thompson	Female	42	Warwick, RI	Marketing Manager	\$78,000/year	Daily — work, online shopping, banking, streaming, and product research
Margaret Grenadine	Female	73	Coventry, RI	Retired	Fixed income	Daily — banking, email, shopping, streaming, and general web research

4. User Stories

US1: Alex Martinez

Profile

- Gender: Male
- Age: 21
- Location: Kingston, Rhode Island
- Occupation: College Student / Part-Time Retail Employee
- Income: \$18,000/year
- Web Use: Daily — schoolwork, shopping, social media, streaming, and technology research

Reason for visiting TechTrade:

Alex's three-year-old laptop has become slow. He wants to know whether he should repair, upgrade, sell, or replace it without spending more money than necessary.

User Story:

As a college student with a limited budget, I want to evaluate whether I should repair, upgrade, sell, or replace my aging laptop so that I can keep a reliable computer without spending more money than necessary.

US2: Sarah Thompson**Profile**

- Gender: Female
- Age: 42
- Location: Warwick, Rhode Island
- Occupation: Marketing Manager
- Income: \$78,000/year
- Web Use: Daily — work, online shopping, banking, streaming, and product research

Reason for visiting TechTrade:

Sarah's five-year-old laptop is experiencing performance problems. She wants to determine whether repairing or upgrading it makes more financial sense than purchasing a replacement.

User Story:

As a busy professional who isn't an expert in computer hardware, I want a simple way to compare repairing, upgrading, selling, or replacing my laptop so that I can make a confident purchasing decision.

US3: Margaret Grenadine

Profile

- Gender: Female
- Age: 73
- Location: Coventry, Rhode Island
- Occupation: Retired
- Income: Fixed income
- Web Use: Daily — banking, email, shopping, streaming, and general web research

Reason for visiting TechTrade:

Margaret's current laptop is several years old and has begun experiencing performance issues. She relies on her laptop for everyday activities but lives on a fixed income and wants to avoid spending money on a new computer if her existing device can be repaired or upgraded.

User Story:

As a retired person living on a fixed income, I want a simple way to understand whether I should repair, upgrade, sell, or replace my laptop so that I can maintain a reliable computer without making an unnecessary expense.

5. Reasons People Would Come to TechTrade

Users may visit TechTrade for several different reasons:

1. **Their current device is running slowly.**
2. **Their device is experiencing technical problems.**
3. **They are considering repairing their device.**
4. **They want to know whether an upgrade would extend the device's useful life.**
5. **They are considering selling their current device.**
6. **They are considering purchasing a replacement device.**
7. **They want to avoid spending more money than necessary.**
8. **They are unsure whether their current device is still valuable.**
9. **They do not understand their device's specifications or capabilities.**
10. **They want a recommendation based on their personal needs and budget.**
11. **They want to research a technology purchase before making a financial decision.**
12. **They want an easy-to-understand alternative to navigating complicated technology marketing and specifications.**

6. User Tasks Assigned to Fictional Visitors

Visitor	Primary Task	Secondary Task
Alex Martinez	Determine whether his three-year-old laptop should be repaired, upgraded, sold, or replaced	Find the most financially responsible option while staying within his limited budget
Sarah Thompson	Compare repairing/upgrading her five-year-old laptop against purchasing a replacement	Determine which option provides the best practical value for her work needs
Margaret Grenadine	Determine whether her aging laptop can be repaired or upgraded instead of replaced	Find a financially manageable solution that allows her to continue using her computer without an unnecessary large expense

7. How These Users Interact With TechTrade

Each visitor would follow the same general process:

Step 1 — Visit the Homepage

The user learns what TechTrade does and selects "**Give a Quote**" to begin the analysis.

Step 2 — Enter Device Information

The user provides information about their current device through text fields and dropdown menus.

Step 3 — Provide Needs and Budget

The user explains what they use the device for and provides information about their financial situation or purchasing budget.

Step 4 — Submit the Analysis

TechTrade analyzes the information provided by the user.

Step 5 — Receive a Recommendation

The user receives a recommendation explaining whether they should:

- Repair
- Upgrade
- Sell
- Replace

The recommendation may also include appropriate links and a downloadable PDF.

Step 6 — Take Action or Restart

The user can follow relevant recommendations or restart the application process to evaluate another device or scenario.

8. Overall Target Audience Summary

TechTrade is designed for everyday technology consumers who want to make better decisions about their existing devices and their money. The audience ranges from young consumers with limited budgets to working professionals and older adults living on fixed incomes.

Although these users have different ages, incomes, occupations, and levels of technical knowledge, they share a common problem: **they need to determine what to do with their technology without making an unnecessary or financially inefficient decision.**

TechTrade addresses this problem by combining information about the user's current device, personal needs, and budget into a simple analysis and recommendation process.

The ultimate goal is to make technology decision-making more intuitive, accessible, and financially responsible for a wide variety of consumers.

9. Purpose & Goals

9.1 What is the primary goal of your website?

The primary goal of TechTrade is to **educate users and help them make more informed decisions about their technology and finances.**

Rather than simply encouraging users to purchase a new device, TechTrade is designed to help users understand whether they should **repair, upgrade, sell, or replace** their current technology. The website will take information about the user's existing device, needs, and budget and use that information to produce a recommendation.

Education is an important part of the experience because many consumers may not fully understand their device's specifications, current value, or the financial implications of replacing it. TechTrade should make this information easier to understand without requiring users to already be technology experts.

For this project, the analysis process will be simulated using hard-coded values for specific devices. In a future, fully developed version of TechTrade, this system could use real-time device information, pricing, specifications, and market data to provide more dynamic recommendations.

9.2 What is the most important action you want users to take on your site?

The most important action I want users to take is to **submit the analysis form and complete the recommendation process.**

The form is the central feature of the website because it allows TechTrade to understand the user's situation and provide a recommendation that is relevant to them.

The goal is not simply to get users to click a button or make a purchase. Instead, users should:

1. Learn about their current technology.
2. Provide information about their device, needs, and budget.
3. Submit the analysis form.
4. Receive and understand their personalized recommendation.
5. Use that information to make a more informed financial decision.

10. Content & Features

10.1 What content and features are essential to help users achieve their goals?

The website will prioritize simplicity while still providing users with enough control to accurately describe their situation.

The most important feature will be the **analysis form**. I want users to have granular control over the information they provide so that the recommendation can be based on their specific circumstances. For the purposes of this project, the available devices and their information will be simulated using hard-coded values rather than a live database.

Critical Pages

Homepage

The homepage will introduce TechTrade, explain the problem it solves, and provide a clear path into the analysis process.

The primary call-to-action will be **"Give a Quote"** or a similar action that takes users directly to the analysis form.

Analysis Input

The Analysis Input page will contain the primary interactive form.

Users will be able to provide information such as:

- Current device
- Device specifications
- Device age
- Device condition
- Performance problems
- Intended usage
- Personal needs
- Budget
- Other relevant circumstances

Dropdowns, text fields, and other appropriate form controls will be used to make the process easy to understand while still providing granular control.

For this project, selecting a device will correspond to hard-coded information about that device so that the recommendation system can be demonstrated without requiring a live product database.

Recommendation Output

After submitting the analysis form, users will receive a recommendation based on the information they entered.

The recommendation page should clearly explain the reasoning behind the recommendation rather than simply telling the user what to do.

The output may include:

- Recommended action
- Explanation of the recommendation
- Relevant device information
- Financial considerations
- Appropriate links
- Downloadable PDF recommendation
- Option to restart the analysis

About

The About section will explain the purpose of TechTrade, why the product was created, and how the service is intended to help consumers make better technology and financial decisions.

10.2 Additional Features

Forgiving Form Functionality

The analysis process should be forgiving and easy to correct.

Users should not feel like they have permanently committed to an option after making a selection. They should be able to easily change previous selections, go back, restart the process, or correct information before submitting the form.

This is especially important because TechTrade is intended for users with different levels of technical knowledge.

Clear Feedback

The website should provide clear feedback when users interact with buttons, forms, or other elements. Users should always have a general understanding of what the website is doing and what action they should take next.

Recommendation Explanation

The recommendation should educate the user instead of functioning as a simple "yes" or "no" answer.

For example, if TechTrade recommends upgrading a laptop instead of replacing it, the user should be able to understand why that recommendation was made and what factors influenced the decision.

PDF Output

The recommendation page will include a PDF download option so users can save or reference their results later.

Restart Function

Users should be able to restart the analysis process easily. This allows someone to evaluate another device or test a different set of circumstances without having to navigate back through the entire website.

11. Homepage

11.1 What will your homepage highlight?

The homepage will focus on **immediately explaining what TechTrade does and allowing users to begin the analysis process.**

The first impression should be straightforward rather than overwhelming. A visitor should be able to understand the purpose of TechTrade within a few seconds.

The homepage should prominently communicate the basic concept:

Should you repair, upgrade, sell, or replace your device?

From there, the page can briefly explain how TechTrade analyzes the user's device, needs, and budget to help them make a more informed decision.

The primary call-to-action will lead directly into the analysis form.

The homepage will also contain an About section that provides additional context for users who want to learn more before beginning the process.

11. Look & Feel

11.1 What websites inspire your design?

ChatGPT

The primary design inspiration for TechTrade is **ChatGPT**.

I want TechTrade to closely follow the minimalist and user-friendly philosophy of ChatGPT rather than creating a website filled with unnecessary visual elements.

Specifically, I like the way ChatGPT focuses attention on the primary interaction while keeping navigation straightforward. The interface does not require users to understand a complicated system before they can begin using it.

TechTrade will take inspiration from this approach by prioritizing:

- Minimalist design
- Straightforward navigation
- Clear calls-to-action
- Simple information hierarchy
- Intuitive interactions
- Generous spacing
- Responsive design
- Subtle animations
- Fast-feeling interactions

The goal is for the website to feel approachable to both technically experienced users and people who may be less comfortable with technology.

Additional Inspiration: Apple

Another source of inspiration is **Apple's general approach to product presentation and interface design**.

I like the emphasis on clean layouts, strong visual hierarchy, and presenting information in a way that does not overwhelm the user. TechTrade can take inspiration from this approach without becoming a product storefront.

The website should make complex technology information feel approachable and understandable.

12. Overall Style

The overall style of TechTrade will be **minimalist, modern, professional, and approachable**.

The design should avoid unnecessary visual clutter and instead focus on the user's task.

The website should feel:

- Clean
- Modern
- Trustworthy
- Simple
- Intelligent
- Approachable
- Responsive
- Professional

The design should also feel slightly interactive without becoming distracting. Animations should primarily communicate feedback, transitions, or changes in state rather than existing purely for decoration.

13. Color, Fonts & Imagery

Color

The color palette should be relatively restrained and minimalist.

I would prefer a neutral foundation consisting primarily of:

- White or light backgrounds
- Dark text
- Neutral gray elements
- A single accent color for important actions and interactive elements

The accent color can be used for buttons, links, selected options, progress indicators, and other interactive elements.

The overall goal is to make the interface feel similar to a modern productivity or technology application rather than a traditional retail website.

Fonts

The typography should be clean, modern, and highly readable.

A simple sans-serif font should be used throughout the website. The typography should create a clear hierarchy between:

- Page titles
- Section headings
- Instructions
- Form labels
- Body text
- Buttons
- Recommendation results

Readability is particularly important because TechTrade is intended to educate users and explain potentially complicated technology and financial information.

Imagery

Imagery should be used sparingly.

TechTrade's primary value comes from its analysis and recommendation process, so large amounts of decorative imagery are unnecessary.

When imagery is used, it should support the content rather than distract from it. Device images, simple illustrations, icons, or graphics could be used where they help users understand a device or recommendation.

14. Design Philosophy

The overall design philosophy for TechTrade is:

"Make the complicated feel simple."

The website should not assume that every visitor understands computer hardware, device specifications, or technology pricing.

Instead, TechTrade should guide users through the process while giving them enough control to accurately describe their situation.

The combination of **granular analysis controls, straightforward navigation, forgiving functionality, clear explanations, and subtle animations** should create an experience that feels simple without being overly simplistic.

The ultimate goal is for a user to visit TechTrade with a technology-related question and leave with a better understanding of their options and a recommendation they can use to make a more informed decision.

Design & Prototyping

SDLC Approach & Timeline

SDLC Model: Agile

For the development of TechTrade, I will use an **AGILE Software Development Life Cycle (SDLC)**. AGILE is appropriate for this project because the website is intended to be developed around user feedback and continuously improved throughout the development process.

Rather than completing every stage once and moving on permanently, the development process will allow me to design, build, test, receive feedback, and make improvements throughout the semester.

This approach is especially useful for TechTrade because the website's primary goal is to create an intuitive recommendation experience. Testing the interface with users can reveal areas where the analysis process, navigation, or recommendations need to be simplified or changed.

The overall development process will consist of:

Planning → Design → Prototype → Development → Testing → Feedback → Refinement → Final Product

SDLC Timeline

Estimated Project Duration: September 17, 2026 – December 16, 2026

Phase	Dates	Primary Activities	Milestone
1. Planning & Requirements	Sept. 17 – Sept. 27	Define problem, goals, target audience, user stories, website requirements, and project scope	Requirements finalized
2. Information Architecture & UX Design	Sept. 28 – Oct. 9	Finalize site map, navigation, page structure, user flow, and analysis process	Site structure finalized
3. Visual Design & Prototyping	Oct. 10 – Oct. 23	Develop wireframes, visual direction, typography, colors, components, and interactive prototype	Prototype completed
4. Initial Development	Oct. 24 – Nov. 6	Build homepage, navigation, About section, analysis form, and basic responsive layout	Functional website foundation
5. Analysis & Recommendation Development	Nov. 7 – Nov. 20	Implement simulated device data, form logic, recommendation outputs, and PDF/output functionality	Recommendation process functional
6. Testing & User Feedback	Nov. 21 – Nov. 30	Test navigation, forms, responsiveness, usability, and recommendation flow; collect feedback	Testing and feedback completed
7. Refinement & Iteration	Dec. 1 – Dec. 9	Fix bugs, improve UI/UX, refine animations, simplify confusing interactions, and incorporate feedback	Final revisions completed
8. Final Testing & Presentation Preparation	Dec. 10 – Dec. 15	Perform final quality assurance, verify all features, polish presentation, and prepare final demonstration	Final product ready
9. Final Submission	Dec. 16	Submit/present completed TechTrade project	Project complete

Visual Timeline Estimation

SEPTEMBER

- Sept. 17–27
PLANNING & REQUIREMENTS
 - └─ Goals • Audience • User Stories • Requirements
- Sept. 28–Oct. 9
INFORMATION ARCHITECTURE & UX
 - └─ Site Map • Navigation • User Flow

OCTOBER

- Oct. 10–23
VISUAL DESIGN & PROTOTYPING
 - └─ Wireframes • Colors • Typography • Components
- Oct. 24–Nov. 6
INITIAL DEVELOPMENT
 - └─ Homepage • Navigation • About • Analysis Form

NOVEMBER

- Nov. 7–20
ANALYSIS & RECOMMENDATION DEVELOPMENT
 - └─ Device Data • Form Logic • Recommendations • PDF
- Nov. 21–30
TESTING & USER FEEDBACK
 - └─ Usability • Responsiveness • Forms • Feedback

DECEMBER

- Dec. 1–9
REFINEMENT & ITERATION
 - └─ Bug Fixes • UX Improvements • Animations
- Dec. 10–15
FINAL TESTING & PRESENTATION
 - └─ QA • Polish • Final Demonstration
- DEC. 16
★ FINAL SUBMISSION ★

Agile Development Cycle

Although the timeline is divided into phases, development will remain iterative. After the initial prototype and development work, testing and user feedback will be used to identify areas for improvement.

The iterative process will follow this cycle:

Design → Build → Test → Gather Feedback → Improve → Repeat

For example, if users have difficulty understanding one of the analysis form questions, the form can be redesigned and tested again rather than waiting until the end of the semester to address the problem.

This approach supports the overall goal of TechTrade: creating a **user-friendly and purpose-built decision-making tool** rather than simply creating a website that technically functions.

Major Project Milestones

Milestone 1 — Requirements Complete

September 27

The problem, target audience, user stories, website goals, and essential requirements will be established.

Milestone 2 — UX Structure Complete

October 9

The site map, navigation, and overall user flow will be finalized.

Milestone 3 — Prototype Complete

October 23

A visual and interactive prototype will demonstrate how users are expected to navigate and interact with TechTrade.

Milestone 4 — Functional Website Foundation

November 6

The primary pages and analysis form will be functional.

Milestone 5 — Recommendation System Complete

November 20

The simulated analysis process, hard-coded device information, and recommendation output will be functional.

Milestone 6 — Testing Complete

November 30

The website will be tested for functionality, usability, responsiveness, and overall user experience.

Milestone 7 — Final Version

December 9

Feedback will be incorporated and major bugs or usability issues will be resolved.

Final Milestone — Submission

December 16

TechTrade will be finalized and prepared for submission and/or presentation.

Development Strategy

The primary focus throughout the semester will be on creating a **simple, intuitive, and forgiving user experience**. The analysis form will receive particular attention because it is the most important feature of TechTrade.

For the purposes of this project, device information and recommendations will be simulated using hard-coded values. This allows the project to demonstrate the intended analysis and recommendation experience without requiring the development of a complete live product database.

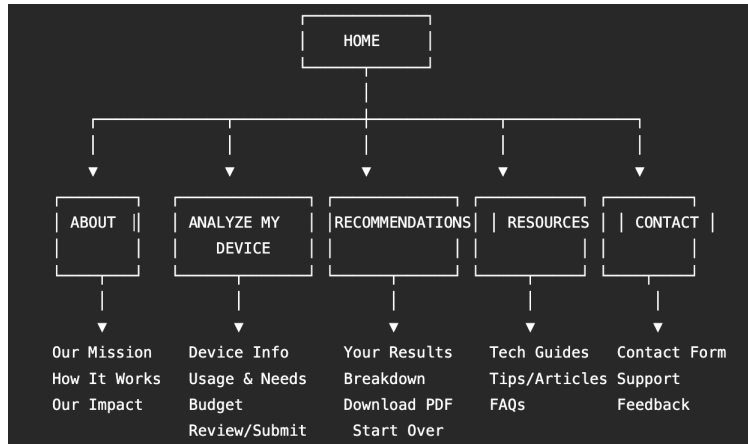
The visual design will follow a minimalist approach inspired by modern productivity and technology interfaces. Navigation will remain straightforward, interactions will provide clear feedback, and animations will be used primarily to make transitions and interactions feel responsive rather than as unnecessary decoration.

By the end of the semester, the goal is to have a functional prototype that demonstrates the complete TechTrade experience:

Enter TechTrade → Learn About the Service → Provide Device & User Information → Submit Analysis → Receive Recommendation → Review/Save Results

Site Map

The TechTrade site map is designed around the primary user journey: learning about the service, analyzing a device, and receiving a recommendation.



Primary Navigation

The navigation bar will remain consistent across the website and provide access to:

- Home
- About
- Analyze My Device
- Resources
- Contact

The **Analyze My Device** option will remain visually prominent because it represents the primary action TechTrade wants users to take.

The Recommendation page will primarily be reached by completing the analysis process rather than functioning as a separate destination that users need to navigate to manually.

Wireframes

The wireframes will focus on three core pages:

1. Homepage
2. Analysis Form
3. Recommendation Output

These pages represent the most important portion of the TechTrade user experience and demonstrate the complete process from entering the website to receiving a recommendation.

Homepage Wireframe

The homepage will provide a simple introduction to TechTrade and immediately communicate the purpose of the website.

Navigation

The top navigation bar will contain:

TechTrade Logo | Home | About | Analyze | Resources | Contact

A primary **Analyze My Device** button will be visible within the page's main content.

Hero Section

The hero section will immediately answer the user's primary question:

"Should you repair, upgrade, sell, or replace your device?"

A short explanation will communicate that TechTrade analyzes the user's device, needs, and budget to help determine the most appropriate option.

The primary call-to-action will be:

Analyze My Device →

Supporting Content

Below the hero section, three benefits will be highlighted:

- **Get a Personalized Recommendation**
- **Save Money**
- **Make a More Informed Choice**

A short "How It Works" section will then explain the process in three steps:

1. Tell Us About Your Device
2. We Analyze Your Situation
3. Get Your Recommendation

The page will end with another clear call-to-action allowing the user to begin the analysis.

Analysis Form Wireframe

The Analysis Form is the most important interactive portion of the website.

The design will provide users with **granular control** over the information they provide while keeping the interface approachable.

A progress indicator will show the user's position in the process:

Device → Usage → Budget → Review

Form Sections

The first section will collect information about the user's current device:

- Device type
- Brand
- Model
- Age
- Condition
- Primary issues

Additional sections will collect:

- How the user uses the device
- Performance requirements
- Budget
- Preferences
- Other relevant circumstances

Navigation

Users will be able to move backward and forward through the form.

Buttons such as:

← **Back**

and

Next →

will allow users to correct information without losing their progress.

This forgiving functionality is particularly important because TechTrade is designed for users with different levels of technical knowledge.

Project Implementation

For this semester project, the device information and recommendation logic will be simulated using **hard-coded values for selected devices**.

The interface will therefore demonstrate how the real TechTrade application would work without requiring a live product database.

Recommendation Output Wireframe

The Recommendation page will present the results of the analysis in a clear and easy-to-understand format.

The page will begin with:

Your Recommendation

followed by a concise statement identifying the recommended action.

Possible recommendation categories include:

- Repair
- Upgrade
- Sell
- Replace

The recommendation will not simply tell users what option to choose. It will explain **why** that recommendation was generated.

Recommendation Breakdown

The page may display factors such as:

- Device age
- Current performance
- User needs
- Upgradeability
- Estimated device value
- Estimated repair costs
- Replacement considerations

The user will therefore be able to understand the reasoning behind the recommendation and become more educated about their options.

Additional Actions

The recommendation page will provide:

- **Download PDF**
- **Start Over**
- **Explore Resources**

This allows users to save their results, analyze another device, or continue learning about technology-related decisions.

Accessibility Plan

Accessibility will be considered from the beginning of the design and development process rather than added after the website is completed.

TechTrade is intended for a broad audience ranging from young college students to older adults, including people who may have different levels of technical knowledge or physical and cognitive accessibility needs.

The goal is to make the website usable for as many people as possible.

Visual Accessibility

The website will include:

- Strong color contrast between text and backgrounds
- Readable font sizes
- Clear visual hierarchy
- Proper heading structure
- Descriptive alt text for meaningful images
- Responsive layouts
- Text that can be resized without breaking the interface
- Icons that are accompanied by text when necessary
- Information that is not communicated through color alone

The minimalist design will also help reduce unnecessary visual clutter.

Accessibility Testing

Accessibility will be tested throughout development rather than only immediately before submission.

The primary tools used to audit TechTrade will include:

Browser Developer Tools

Browser developer tools will be used to inspect page structure, keyboard focus, responsive behavior, and other implementation details.

Manual Keyboard Testing

The site will also be tested manually using only a keyboard to verify that users can navigate the website, interact with the form, submit information, and access the recommendation without requiring a mouse.

Final Design Direction

The final TechTrade design will combine a **minimalist visual style, straightforward navigation, granular analysis controls, forgiving interactions, and clear recommendations.**

The three core pages—Homepage, Analysis Form, and Recommendation Output—will create a simple and predictable user journey:

Learn → Analyze → Understand → Decide

The visual design will remain intentionally clean so that the technology and financial information does not overwhelm the user. Subtle, responsive animations can be used to make transitions and interactions feel fast and polished without distracting from the primary task.

Ultimately, the design should make TechTrade feel less like a traditional technology retailer and more like an intelligent, approachable tool that helps users understand their options before they spend their money.